Title: Multi-channel 12 GHz microwave synthesizer

Item 1. Multi-channel microwave synthesizer

Item 1 Quantity: 1

1.1 Three (3) independent channels having the capability to generate sinusoidal output from 10 MHz to 12 GHz with the following attributes

1.1.1 SMA connectors, 50 Ohm nominal load
1.1.2 Output power ranges from -10 to 15 dBm
1.1.3 Frequency step size at most 0.001 Hz
1.1.4 Frequency switching speed at most 500 microseconds
1.1.5 Amplitude switching speed at most 500 microseconds
1.1.6 Harmonic amplitude relative to 0 dBm carrier < -25 dBc over 1 – 10 GHz frequency range
1.1.7 SSB Phase noise at 6 GHz less than -90 dBc/Hz @ 100 Hz offset
1.1.8 SSB Phase noise at 6 GHz less than -110 dBc/Hz @ 10 kHz offset
1.1.9 SSB Phase noise at 12 GHz less than -85 dBc/Hz @ 100 Hz offset
1.1.10 SSB Phase noise at 12 GHz less than -95 dBc/Hz @ 10 kHz offset
1.1.11 Ability to perform list-based scans of frequency and amplitude

1.2 Two (2) independent channels having capability to generate sinusoidal output from 10 MHz to 6 GHz with the following attributes (these channels could also be 12 GHz channels, but this capability is not necessary)

1.2.1 SMA connectors, 50 Ohm nominal load
1.2.2 Output power ranges from -10 to 15 dBm
1.2.3 Frequency step size at most 0.001 Hz
1.2.4 Frequency switching speed at most 500 microseconds
1.2.5 Amplitude switching speed at most 500 microseconds
1.2.6 Harmonic amplitude relative to 0 dBm carrier < -25 dBc over 1 – 10 GHz frequency range
1.2.7 SSB Phase noise at 6 GHz less than -90 dBc/Hz @ 100 Hz offset
1.2.8 SSB Phase noise at 6 GHz less than -110 dBc/Hz @ 10 kHz offset
1.2.9 Ability to perform list-based scans of frequency and amplitude
1.2.10 Dedicated analog modulation port for frequency control with 100 kHz deviation and 20 kHz of modulation bandwidth using a +/- 1V signal

1.3 Synthesizer should have a 10/100 MHz reference input and a 100 MHz reference output
1.4 Synthesizer should be power by standard US 110 V AC and consume less than 100 W
1.5 Synthesizer should operate under normal laboratory conditions (25 C)
1.6 Synthesizer should have the ability to program via USB and/or SPI (or equivalent digital protocols)
1.7 Synthesizer chassis should be in a 19” rack mountable chassis less than 3U high, doesn’t necessarily need 19” rack mounting hardware

1.8 Synthesizer should have a calibration certification

1.9 Synthesizer should controllable via a computer through a graphical user interface, a front panel interface is not necessary for this instrument